

Vincent (Dam Duc Huy) Ngo

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EDUCATION

University of Alberta

B.Sc. Computer Engineering, Co-op - GPA: 3.9/4.0

Edmonton, AB

Expected 2029

HARDWARE, VALIDATION & SYSTEMS EXPERIENCE

Electrical and Control Engineering Intern

May 2026 – Aug. 2026

Red Deer Polytechnic - CIM-TAC / Energy Innovation Centre

Red Deer, AB

- Designed, integrated, and debugged embedded platforms spanning **ESP32-S3/C6, MSP430FR6043, PIC18F24Q10, mixed-signal PCBs, sensors, power electronics, and industrial controls**, using oscilloscopes, logic analyzers, DMMs, protocol tools, and power profiling for root-cause analysis.
- Engineered a **65 mm x 18 mm, 4-layer mixed-signal PCB** for a constrained biomechanical telemetry system, integrating IMU and force sensing, ideal-diode power-path management, low-noise analog rails, and 50 Hz BLE telemetry; diagnosed thermal IMU bias and reverse-current power faults during bring-up.
- Commissioned a **230 VAC 3-phase, 4.5 kW servo system** with dual-channel Safe Torque Off, EMI filtering, isolated analog/digital interfaces, RS-485/Modbus, and 4-20 mA instrumentation; performed isolation checks, live measurements, regenerative-braking debug, and closed-loop system validation.
- Developed an in-circuit **Spy-Bi-Wire ROM bootloader** on ESP32-C6 for MSP430FR6043 and traced target brownout/boot failures to parasitic back-powering across shared GPIOs, resolving the issue through High-Z sequencing and controlled power gating verified with a Nordic PPK2.
- Architected dual-core FreeRTOS firmware for a sensing platform and reduced average current draw **70.4% (85.71 mA to 25.33 mA)**; diagnosed a queue-related task shutdown soft-lock and validated interrupt-driven power-state behavior across hardware revisions.
- Built repeatable firmware/test workflows across Windows and Linux by wrapping Microchip XC8, TI toolchains, and **arduino-cli**; migrated BLE to NimBLE to reduce binary size by approximately **600 KB (over 35%)** and support dual-image OTA validation.

Flight Software & Embedded Systems Team Member

Sept. 2024 – Apr. 2025

AlbertaSat - Ex-Alta 3 CubeSat

Edmonton, AB

- Performed hardware-in-the-loop integration and ISO 8 clean-room qualification for a 3U CubeSat OBC, debugging **UART, SPI, I2C, and CAN** interfaces among ADCS, EPS, and payload subsystems with oscilloscopes and logic analyzers.
- Contributed to an OBC stack spanning **embedded Linux (Yocto/PetaLinux), RTOS, Rust flight software, BIOS/boot configuration, and VHDL/Verilog FPGA logic**; supported fault-tolerant boot, watchdog, telemetry, and hardware-software integration workflows.
- Authored version-controlled test procedures, acceptance test reports, and verification documentation while tracing communication race conditions and signal-integrity issues across integrated spacecraft subsystems.

HARDWARE & SYSTEMS PROJECTS

Bare-Metal Real-Time Audio Spectrum Visualizer | STM32, C, CMSIS-DSP, DMA, PWM

2025

- Built a register-level STM32F103C8 system using ADC, DMA, timers, and a **1024-point CMSIS-DSP FFT**; profiled a 28 ms bottleneck with GDB and reduced processing to **15 ms**, achieving 17 ms end-to-end latency within a 20 KB SRAM budget.
- Validated ADC-to-SRAM DMA transfers and nanosecond-accurate 800 kHz WS2812B timer-PWM waveforms using a logic analyzer and oscilloscope, confirming timing margins and non-blocking operation under continuous acquisition.

HARDWARE & SYSTEMS PROJECTS

NEST Wearable Health Monitor | *ESP32, ESP-IDF, FreeRTOS, KiCad* 2025

- Designed and assembled a compact 2-layer mixed-signal PCB integrating MAX30102, MPU6500, ENS160, and AHT21 sensors; performed hardware bring-up, 3.3 V rail validation, and I2C/SPI signal-integrity debugging with oscilloscopes and multimeters.
- Architected dual-core FreeRTOS firmware with SPI DMA and inter-core synchronization, isolating high-rate sensor acquisition from display and Wi-Fi tasks and resolving display-induced task starvation during system integration.

High-Throughput Serial & BLE Telemetry Engine | *C#, .NET 8, System.IO.Pipelines* 2026

- Built a multithreaded diagnostic telemetry engine for serial/BLE hardware, implementing resilient packet framing, checksum validation, mixed-endian decoding, and bounded background pipelines for sustained logging without UI stalls.
- Used the tooling to investigate high-rate embedded telemetry and disconnect behavior, eliminating buffer backpressure and packet-loss failure modes in host-side validation workflows.

Automated Embedded Build & Test Deployment Pipeline | *Bash, PowerShell, WSL2, XC8* 2026

- Automated multi-target firmware builds, binary staging, and in-circuit/OTA deployment around `arduino-cli` and native Linux `xc8-cc`, cutting build-and-flash overhead by **over 60%** and improving repeatability across test hardware.
- Resolved Windows/WSL path and vendor-toolchain interoperability failures through POSIX wrappers and environment automation, enabling reproducible diagnostics and firmware validation outside IDE-dependent workflows.

Biomechanical Telemetry Validation Dashboard | *Flutter, Dart, BLE* 2026

- Developed a cross-platform diagnostic dashboard that decoded 180-byte BLE sensor frames, graphed multi-axis force and motion telemetry at 60 FPS, and logged synchronized sessions for hardware/firmware test analysis.
- Eliminated UI frame drops and lifecycle-related memory leaks by moving binary parsing outside rendering widgets and isolating high-rate graph repaints, improving reliability during sustained system validation runs.

TECHNICAL SKILLS

Hardware Validation & Lab: hardware bring-up, board/system integration, root-cause analysis, oscilloscopes, logic analyzers, DMMs, protocol analysis, Nordic PPK2, clean-room qualification, SMT assembly/rework

Programming & Scripting: C, C++, Embedded C, Python, Rust, C#, Bash, PowerShell; multithreaded systems, CLI tools, Linux/POSIX environments

Embedded & Real-Time: FreeRTOS, ESP-IDF, bare-metal ARM/CMSIS, GPIO, ADC, DMA, timers, ISRs, peripheral drivers, bootloaders, firmware debugging

MCUs & Compute: STM32 (ARM Cortex-M), ESP32-S3/C6, MSP430FR6043, PIC18F24Q10, FPGA/SoC platforms, RISC-V MCU exposure

Embedded Linux & Toolchains: Yocto Project, PetaLinux, Buildroot, WSL2, GNU Arm, Microchip XC8, TI CCS/MSP430 CGT, Git, GDB, cross-compilation

Digital & FPGA: VHDL, Verilog, Vivado, FPGA digital logic, FSMs, digital systems; exposure to hardware-accelerated telemetry logic

Protocols & Interfaces: UART/USART, SPI, I2C, CAN, BLE, Wi-Fi, TCP/IP, RS-485/Modbus RTU, 4-20 mA, JTAG/SWD, Spy-Bi-Wire

PCB & Signal Processing: KiCad, Altium Designer, mixed-signal PCB design, DFM/DFA, CMSIS-DSP, FFT, real-time signal processing

AWARDS & SCHOLARSHIPS

IEEE Northern Canada Section Scholarship 2026

The Albert Julian Hrapko Memorial Scholarship in Computer Engineering 2025

The Robert Stollery Scholarship in Engineering 2024–2025

Third Place - Natignite 2025 Hackathon 2025